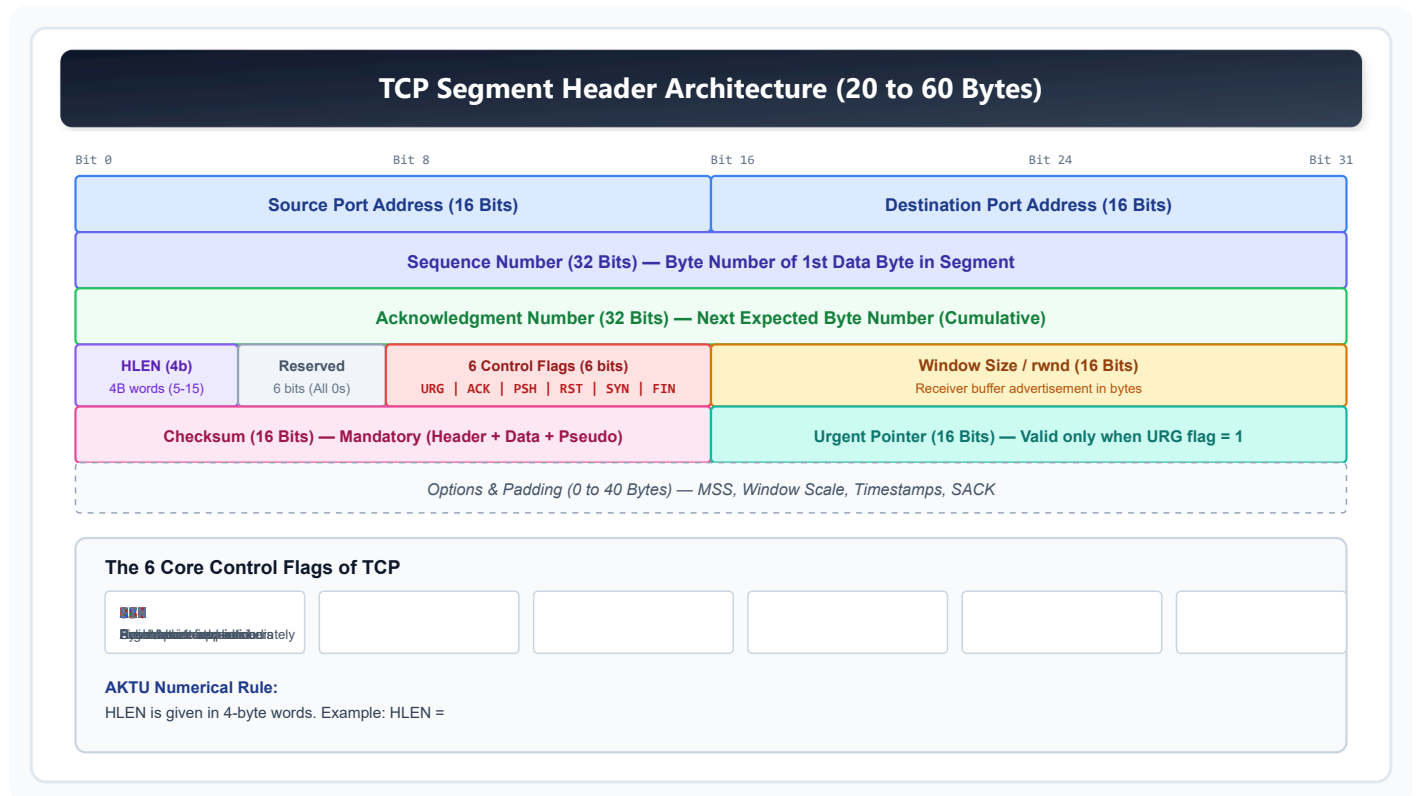


Module 02: TCP Foundations and Segment Header Format

Unit 4: Transport Layer

TCP Header Architecture

Byte-Stream Service, 20-60 Byte Header Field Specifications, 6 Control Flags, Window Advertisement, and Solved AKTU Hex Dump Numericals.



5 $\implies 5 \times 4 = 20$ ext{ Bytes}\$. Maximum HLEN = 15 $\implies 15 \times 4 = 60$ ext{ Bytes}\$. Sequence numbers consume 1 sequence count per SYN and FIN flag even if carrying 0 data bytes!

1. 14 Header Fields Summary

HLEN Equation: Header Length in Bytes = HLEN (4 bits) * 4 (Min = 20B, Max = 60B)

Control Flags (6 bits): URG | ACK | PSH | RST | SYN | FIN

Window Size (16 bits): rwnd buffer advertisement (Max standard = 65,535 Bytes)

2. Solved AKTU Hex Dump Key Findings

Field	Hex Chunk	Decimal / Meaning
Source Port	05 32	1330 (Ephemeral Client Port)
Destination Port	00 17	23 (Telnet Server)
Sequence Number	00 00 00 01	1 (Initial Sequence Number)
Header Length	5 (High nibble of byte 12)	5 * 4 = 20 Bytes (No options)
Active Flags	02 (Byte 13)	SYN = 1 (Connection Request)
Window Size	07 FF	2047 Bytes advertised